Binning

1. Definition

Binning, also called *discretization* or *bucketing*, consists in transforming a continuous numerical variable into a discrete one by grouping its values into intervals called *bins*. For instance, a variable such as age can be converted into categories like "young", "adult", and "senior". This is a common preprocessing technique in machine learning and statistics.

2. Why Use Binning?

Binning is useful for several reasons:

Capturing non-linear effects: some relationships are not well modeled by a linear trend, while bins can represent threshold effects.

Reducing noise: close numerical values are grouped together, which makes small fluctuations less important.

Improving interpretability: categories such as "low", "medium", and "high" are often easier to explain than raw numerical values.

Limiting the influence of outliers: extreme observations are absorbed into boundary bins.

Adapting data to some models: certain methods work naturally with categorical variables or benefit from simplified inputs.

3. Main Types of Binning

Equal-width binning. The interval $[x_{\min}, x_{\max}]$ is divided into k subintervals of identical width:

$$w = \frac{x_{\max} - x_{\min}}{k}.$$

A value x is assigned to the bin

$$b(x) = \min\left(\left\lfloor \frac{x - x_{\min}}{w} \right\rfloor, k-1\right).$$

This approach is simple and easy to interpret, but it may perform poorly on skewed data, since some bins can be almost empty while others contain many observations.

Equal-frequency binning. Here, the bin boundaries are chosen so that each bin contains approximately the same number of observations. In practice, these boundaries are defined using quantiles. This method is often more suitable for skewed distributions, although the bin widths are no longer equal.

Custom binning. The boundaries are chosen using domain knowledge. For example, age groups, BMI classes, or income brackets may be defined according to meaningful real-world thresholds. This often gives the most interpretable categories.

3.1 Important Practical Issues

Several points must be handled carefully:

- the maximum value should belong to the last bin;
- new observations outside the training range may need to be clipped or assigned to a special bin;
- if many observations share the same value, exact equal-frequency binning may be impossible;
- equal-width binning may create empty bins in sparse regions.

4. Advantages and Limitations

Binning can improve robustness, interpretability, and feature engineering, but it also implies a loss of information: two different values that fall in the same bin become indistinguishable. Hence, there is a trade-off between simplicity and precision. Too few bins may lead to underfitting, while too many bins reduce the benefit of discretization.

5. Typical Applications

Binning is widely used in practice, for example in credit scoring, medical diagnosis, histogram construction, risk segmentation, privacy-preserving data analysis, feature engineering for machine learning models.

6. Python Code

```python
def binning(values, num_bins):
    min_val = min(values)
    max_val = max(values)
    w = (max_val - min_val) / num_bins
    ans = [0] * len(values)

    if w == 0:
        return ans

    for i in range(len(values)):
        ans[i] = min(int((values[i] - min_val) / w), num_bins - 1)

    return ans
```